

ABDUL MUQTADIR

(+92) 333 454 7518 | Muqtadir0711@gmail.com | LinkedIn | github.com/muqtadir2777 | muqtadir2777.github.io |

Risalpur, Pakistan

EDUCATION

National University of Sciences and Technology (NUST)

Sep 2023 – Present

Bachelor of Engineering in Avionics Engineering; CGPA: 3.73 / 4.00 (Semester 7)

Risalpur, Pakistan

- **Academic Honors:** Maintained Dean's Honor List standing across 7 consecutive semesters.
- **Relevant Coursework:** RF & Microwave Engineering, Antenna & Wave Propagation, Digital Signal Processing (DSP), Control Systems, Embedded Systems, Microprocessors, Radar Systems, Electronic Warfare, Telemetry.

The Universal Millennium College

Aug 2021 – May 2023

Cambridge International A-Levels — Mathematics, Physics, Chemistry (3 A Grades)

Pakistan

TECHNICAL SKILLS

RF, Microwave & Antennas: CST Studio Suite 2025 (3D Full-Wave EM), Keysight ADS, Keysight LineCalc, S-Parameter Analysis ((S_{11}, S_{21})), CPW & Microstrip Design, Vivaldi Exponential Slot Antennas, Conformal Arrays.

Communications & SDR: ADALM-Pluto Software-Defined Radio, 915 MHz FM Telemetry, Quadrature Demodulation, Digital IIR/FIR Filter Design, Noise Suppression.

Control Systems & Dynamics: Closed-Loop Digital PID Control, Linear Quadratic Regulator (LQR), State-Space Modeling, IMU 6-DOF Sensor Fusion (Kalman & Complementary Filters), Lagrangian Dynamics.

Programming & Software: C/C++, Embedded C, Python (NumPy, SciPy, Pandas, Matplotlib), MATLAB / Simulink, G-Code, Linux/Bash, Git.

Hardware & CAD Prototyping: ESP32, STM32, Arduino Uno, MPU6050, BTS7960 43A Drivers, Stepper Motors, CATIA V5, AutoCAD, KiCAD, Proteus, Oscilloscope Diagnostics.

ENGINEERING PROJECTS

Super Wideband (3–18 GHz) Conformal Vivaldi Antenna

Aug 2026

Lead Designer | CST Studio Suite 2025, Rogers RT6035HTC, CPW Transition

NUST CAE

- Designed and numerically simulated an ultra-compact planar and conformal Vivaldi antenna achieving a 6:1 impedance bandwidth ratio ($(S_{11} < -10\text{ dB})$) from 3.0 to 18.0 GHz for UAV Electronic Warfare (EW) direction-finding.
- Synthesized a single-layer Co-Planar Waveguide (CPW) feed transition and formulated an analytical exponential taper profile ($(R = 0.16\text{ mm}^{-1})$) with 4 curved ground slits to suppress destructive higher-order cross-polarization modes.
- Modeled aerodynamic conformal cylindrical curvature matching UAV fuselage contours and evaluated mutual coupling isolation ($(S_{21} < -25\text{ dB})$).

Hardware-in-the-Loop (HIL) UAV Altitude Control & SDR Telemetry

Jul 2026

Systems & Comms Engineer | ADALM-Pluto SDR, Python, PID Autopilot, 915 MHz FM

NUST CAE

- Constructed a real-time HIL telemetry framework coupling a simulated 2nd-order aircraft dynamics plant with a closed-loop PID altitude autopilot.
- Modulated continuous telemetry states onto a 915 MHz carrier using frequency modulation (FM) and streamed I/Q frames via an ADALM-Pluto SDR.
- Implemented receiver DSP pipeline performing quadrature demodulation and single-pole IIR noise filtering to reconstruct the control feedback loop under simulated AWGN conditions.

2.4 GHz 5th-Order Chebyshev Stepped-Impedance Low-Pass Filter

Jun 2026

RF Designer | Keysight ADS, Keysight LineCalc, CST 3D EM, FR4 Microstrip

NUST CAE

- Synthesized a distributed stepped-impedance microstrip low-pass filter at a 2.4 GHz cutoff frequency for avionics ISM-band harmonic suppression.
- Executed Monte Carlo statistical yield analysis to model substrate permittivity variations ($\epsilon_r = 4.4 \pm 0.1$) and trace width tolerances; verified full 3D S-parameters in CST Studio Suite.

Two-Wheeled Inverted Pendulum Robot: LQR vs. Cascaded PID

May 2026

Control Systems Engineer | MATLAB/Simulink, C++ Firmware, MPU6050 IMU

NUST CAE

- Derived non-linear Lagrangian equations of motion and linearized state-space matrices around the unstable vertical equilibrium.
- Demonstrated performance degradation of state-feedback LQR under unobserved linear cart states $(x, \dot{x})$; engineered and tuned a physical cascaded PID firmware with anti-windup clamping and complementary sensor filtering.

CNC 2D Handwriting Machine & Vector Plotter

Dec 2025

Mechatronics Designer | Arduino Uno, GRBL v1.1, A4988, NEMA 14, CATIA V5

Personal Project

- Designed dual-axis mechanical gantry in CATIA V5 and calibrated GRBL firmware EEPROM to achieve 0.05 mm positional repeatability for vector G-code execution.

Autonomous Heavy Competition Rover

Apr 2025

Hardware & Power Engineer | ESP32 Dual-Core, Dual BTS7960 43A, 4S2P Li-Ion

Robotics R&D

- Engineered heavy-duty powertrain utilizing dual BTS7960 43A H-bridges, JGA-25-370 geared motors, and custom 16.8V battery management with real-time wireless telemetry.

EXPERIENCE & LEADERSHIP

Freelance Engineering Consultant

Jul 2025 – Present

Independent Contractor — Embedded Systems, Firmware & RF

Remote

- Consulted on microcontroller firmware architecture, I2C/SPI sensor interfaces, and closed-loop control algorithms for international clients.
- Performed schematic reviews, circuit debugging, and component selection for high-reliability embedded prototypes.

Head of Avionics Club

Sep 2024 – Present

College of Aeronautical Engineering, NUST

Risalpur, Pakistan

- Direct technical initiatives, SDR telemetry experiments, and robotics workshops across the department.
- Organize technical bootcamps bridging analytical classroom physics with hands-on hardware laboratory implementation.

A-Level STEM Tutor (Mathematics & Physics)

Nov 2022 – Present

Private Mentorship

Remote & In-Person

- Taught calculus, kinematics, wave mechanics, and electromagnetic theory to 20+ students, helping them achieve A/A* grades.

HONORS & CERTIFICATIONS

- **Dean's Honor List:** Recognized for top academic excellence across 7 consecutive semesters at NUST CAE.
- **IBM Professional Certification:** Data Analysis with Python (NumPy, SciPy, Pandas, Scikit-Learn).